Chapter Six

Categorical Propositions

6.1. The Components of CP

- Key Words
  - Proposition
  - Categorical Proposition
  - Subject term
  - Predicate term
  - Quantifiers
  - Copula
  - Standard – Form Categorical Proposition
  - Boolean Standpoint and Aristotelian Standpoint
  - Existential Import
  - Existential Fallacy
  - Immediate inference
  - Conversion, Obversion, contraposition

•A proposition that relates two classes, or categories is called **categorical proposition**.

Examples:

1. Everything that runs has a leg.
2. Many of today's unemployed have given up on finding work.
3. All ducks swim.
4. Penguins cannot fly.

•The classes in question are denoted respectively by the **subject term** and the **predicate term.**

•The proposition asserts that *either all or part of the class denoted by the subject term is included in or excluded from the class denoted by the predicate term.*

- There are exactly **four** types of CP.
  1. Those that assert that the whole subject class is included in the predicate class,
     All men are liars.
  2. those that assert that part of the subject class is included in the predicate class,
     Some men are liars.
  3. those that assert that the whole subject class is excluded from the predicate class, and
     No men are liars.
  4. those that assert that part of the subject class is excluded from the predicate class.
     Some men are not liars.

- A categorical proposition that expresses these relations with complete clarity is called a **standard-form categorical proposition.**
- A categorical proposition is in standard form if and only if it is a substitution instance of one of the following **four** forms:

  All *S* are *P*.
  No *S* are *P*.
  Some *S* are *P*.
  Some *S* are not *P*.
- The words ***"all," "no,"*** and ***"some"*** are called **quantifiers** because they specify how much of the subject class is included in or excluded from the predicate class.
- "*S*" and "*P*" stand for subject term and predicate term, respectively.
- **'Are'** and **'are not'** are called the **copula.**

Example:

Some universities that emphasize research are not institutions that neglect under graduate education.

This *standard-form categorical proposition* is analyzed as follows:

quantifier: *subject term:* *copula:* *predicate term:*	---------- ---------- ---------- ----------

- "All *S* are not *P*" is *not* a standard form. This form is ambiguous and can be rendered as either "No *S* are *P*" or "Some *S* are not *P*," depending on the content.

6.2. Quality, Quantity, and Distribution

- Quality and quantity are attributes of categorical propositions.

All S are P	Every member of the **S** class is a member of the **P** class. (**S** class is included in the **P** class)
No S are P	No member of the **S** class is the member of the **P** class. (The **S** class is excluded from the **P** class.)
Some S are P	At least one member of the **S** class is a member of the **P** class.
Some S are not P	At least one member of the **S** class is not a member of the **P** class.

- The **quality** of a categorical proposition is either **affirmative** or **negative** depending on whether it *affirms* or *denies* class membership.
- Accordingly,
  - "All *S* are *P*"
  - "Some *S* are *P*" } **affirmative** quality,
  - "No *S* are *P*"
  - "Some *S* are not *P*" } **negative** quality.
  - These are called **affirmative propositions** and **negative propositions,** respectively.

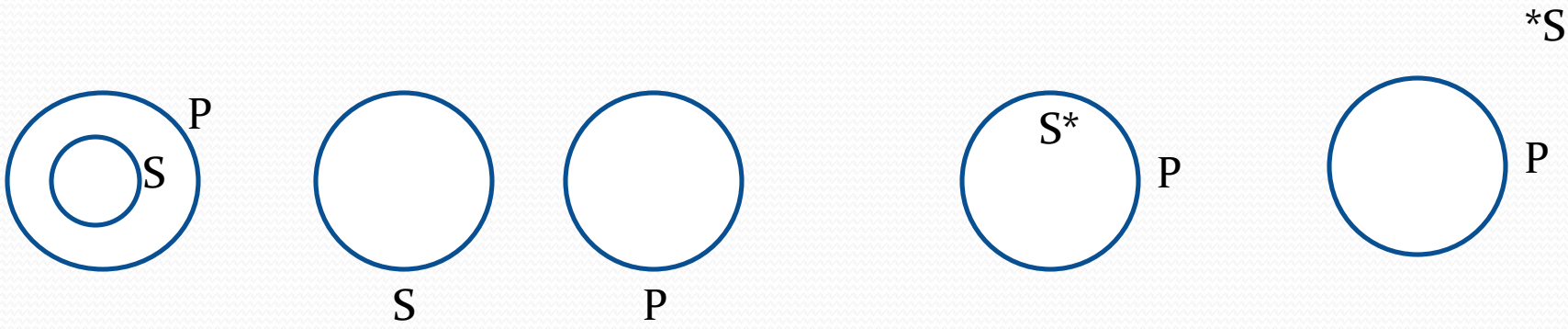
- The **quantity** of a categorical proposition is either **universal** or **particular**, depending on whether the statement makes a claim about *every* member or just *some* member of the class denoted by the subject term.
- "All *S* are *P*" and "No *S* are *P*" each assert something about every member of the *S* class and thus are **universal propositions.**
- "Some *S* are *P*" and "Some *S* are not *P*" assert something about one or more members of the *S* class and hence are **particular propositions.**

•Note that the quantity of a categorical proposition may be determined through mere inspection of the quantifier. “All” and “no” immediately imply universal quantity, while “some” implies particular.

•But categorical propositions have no “qualifier.” In universal propositions the quality is determined by the quantifier, and in particular propositions it is determined by the copula.

- Categorical propositions have commonly been designated by letter names corresponding to the first four vowels of the Roman alphabet: **A, E, I, O**.
- The universal affirmative is called an **A** proposition, the universal negative an **E** proposition, the particular affirmative an **I** proposition, and the particular negative an **O** proposition.

•**Distribution** is an attribute of the *terms* (subject and predicate) of propositions.
•A term is said to be distributed if the proposition makes an assertion about every member of the class denoted by the term; otherwise, it is undistributed.
•If a certain term is *distributed* in a proposition, this simply means that the proposition says something about every member of the class that the term denotes. If a term is *undistributed,* the proposition does not say something about every member of the class.



Proposition	Letter Name	Quantity	Quality	Terms Distributed
All S are P	A	Universal	Affirmative	S
No S are P	E	Universal	Negative	S and P
Some S are P	I	Particular	Affirmative	none
Some S are not P	O	Particular	Negative	P

Class Exercise: Identify the letter name, quantity and quality. Change the quality without changing the quantity, change the quantity without changing the quality, change both the quality and the quantity.

- All governments that bargain with terrorists are governments that encourage terrorism.
- Some symphony orchestras are organizations on the brink of bankruptcy.
- Some affirmative action plans are not programs that result in reverse discrimination.
- No labor strikes are events welcomed by management.

6.3. Venn Diagrams and the Modern Square of Opposition

- There are two possible interpretations of universal propositions **A** and **E**.
- They are known as **Aristotelian** and **Boolean** interpretations.
- Aristotelian standpoint claims that universal propositions imply existence whereas Boolean standpoint claims nothing about existence.
- For Aristotle, universal propositions about existing things have **existential import**.
- For Boole, no universal propositions have existential import.

Existential Import…

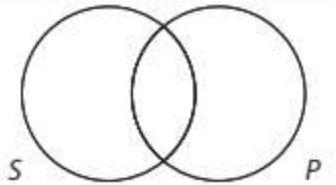
- The Aristotelian standpoint differs from the Boolean standpoint only with regard to universal propositions, **A** and **E**.
- The two standpoints are identical with regard to particular propositions **I** and **O**.
- Both the Aristotelian and the Boolean standpoints recognize that particular propositions make a positive assertion about existence.

Existential import…

- Examples:
- (Aristotle)
  - All students are smart. (Implies existence of students.)
  - All satyrs are vile creatures. (Does not imply existence of satyrs.)
- Boolean
  - All trucks are vehicles. (Does not imply existence of Trucks.)
  - All vampires are monsters. (Does not imply existence of Vampires.)

Venn diagram…

- A **Venn diagram** is an arrangement of overlapping circles in which each circle represents the class denoted by a term in a categorical proposition.



All *S* are *P*. = No members of *S* are outside *P*.

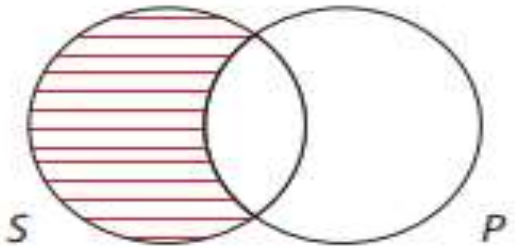
No *S* are *P*. = No members of *S* are inside *P*.

Some *S* are *P*. = At least one *S* exists, and that *S* is a *P*.

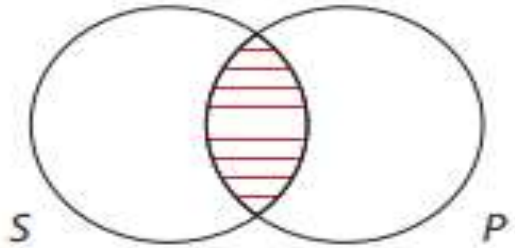
Some *S* are not *P*. = At least one *S* exists, and that *S* is not a *P*

- Two kinds of marks are used:
- **shading** an area and placing an **X** in an area.
- Shading => emptiness,
- X => existence
- If no mark appears in an area, this means that nothing is known about that area; it may contain members or it may be empty.
- Shading is always used to represent the content of universal (**A** and **E**) propositions.
- Placing an X in an area is always used to represent the content of particular (**I** and **O**)propositions.

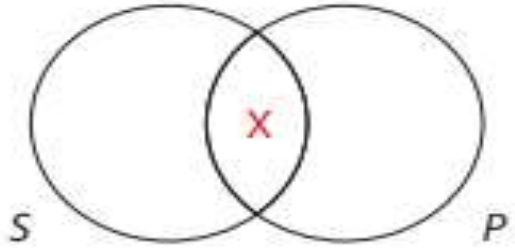
A: All *S* are *P*.



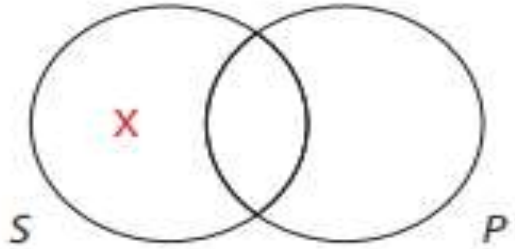
E: No *S* are *P*.



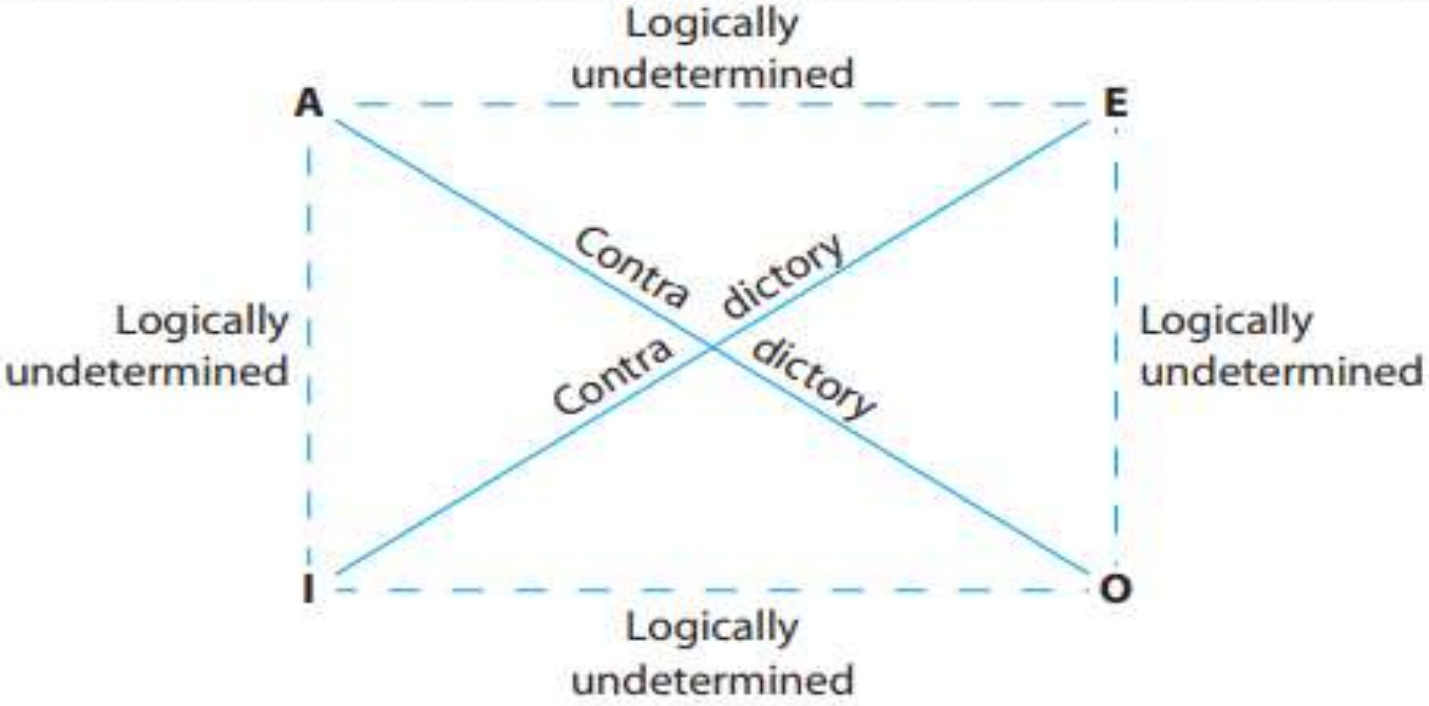
I: Some *S* are *P*.



O: Some *S* are not *P*.



The modern square of opposition



The Modern…

- MSO is used to test the validity of arguments.
- We begin by assuming the premise is true.
- We enter the pertinent truth value in the square.
- If the square indicates that the conclusion is true, the argument is valid, if not the argument is invalid.
- Arguments that are valid from the Boolean standpoint are said to be **unconditionally valid** because they are valid regardless of whether their terms refer to existing things.

Testing Immediate inferences

- Arguments that draw a conclusion from a single premise are called **immediate inferences**.
- Both MSO and Venn diagrams can be used to test the validity of immediate inferences.
- **Examples:**
  1. Some trade spies are not masters at bribery. Therefore, it is false that all trade spies are masters at bribery.
  2. It is false that all meteor showers are common spectacles. Therefore, no meteor showers are common spectacles.
  3. All smartphones are wireless devices. Therefore, some smartphones are wireless devices.

Existential Fallacy

- From the Boolean standpoint, the **existential fallacy** is a formal fallacy that occurs whenever an argument is invalid merely because the premise lacks existential import.
- Such arguments always have a universal premise and a particular conclusion.
- The premise diagram contains shading and the conclusion diagram contains an X.
- If the X in the conclusion diagram is in the same part of the left –hand circle that is unshaded in the premise diagram, then the inference commits the existential fallacy.
- There are exactly **eight** inference forms that commit the existential fallacy.

Existential fallacy….

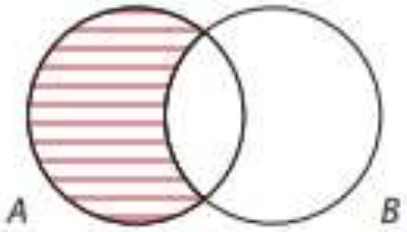
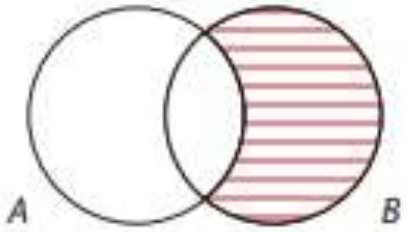
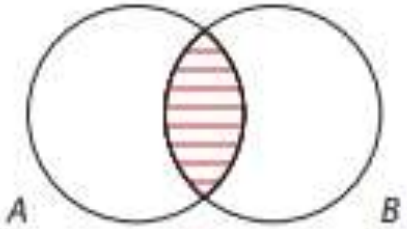
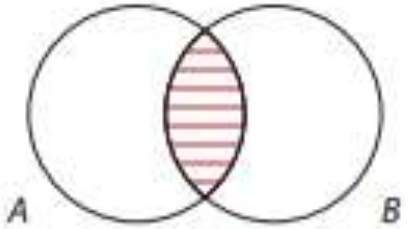
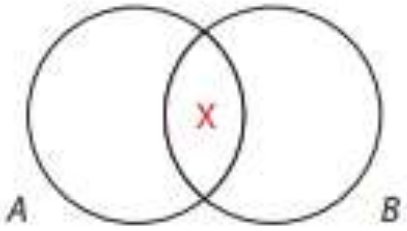
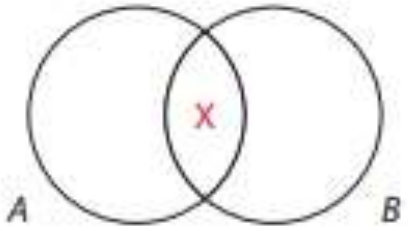
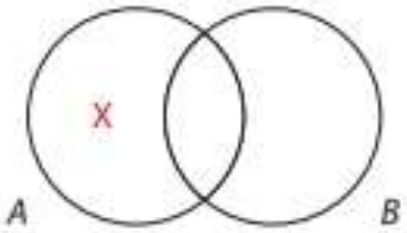
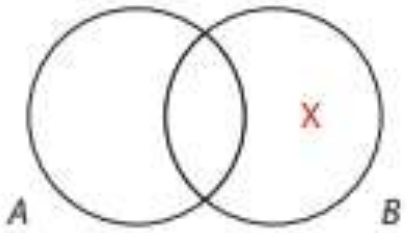
- Examples:
  1. All A are B. Therefore, some A are B.
  2. It is false that some A are not B. Therefore, it is false that no A are B.
  3. No A are B. Therefore, it is false that all A are B.
  4. It's false that some A are B. Therefore, some A are not B.
  5. All A are B. therefore, some A aren't B.

6.4.Conversion,Obversion and Contraposition

- **Conversion**, **obversion**, and **contraposition** are operations that can be performed on a categorical proposition, resulting in a new statement that may or may not have the same meaning and truth value as the original statement. 1.No cats are dogs.=No dogs are cats.2. some cats are animals=some animals are cats. 3. Some students are not philosophers.= some students are non-philosophers.

Conversion

- switching the subject term with the predicate term
  - All S are P → → All P are S
  - No S are P → → No P are S
  - Some S are P → → Some P are S
  - Some S are not P → → Some P are not S

Given statement form		Converse	
All *A* are *B*.		All *B* are *A*.	
No *A* are *B*.		No *B* are *A*.	
Some *A* are *B*.		Some *B* are *A*.	
Some *A* are not *B*.		Some *B* are not *A*.	

- **E and I** statements are identical to that of their converse.
- **E and I** statements and their converse are logically equivalent,
- **E and I** statement and their converse always have the same truth value (and the same meaning).

- The diagram for the **A and O** statements are not identical to the diagram for their converse.
- Also, these pairs of diagrams are not the exact opposite of each other, as is the case with contradictory statements.
- This means that **A and O** statements and their converse are logically unrelated as to truth value.
- In other words, converting an **A** or **O** statement gives a new statement whose truth value is logically undetermined in relation to the given statement.

Examples:

1. All cats are animals. Therefore, all animals are cats.
2. No cats are animals. Therefore, no animals are cats.
3. Some cats are animals. Therefore, some animals are cats.
4. Some animals are not dogs. Therefore, some dogs are not animals.

A and **O** commits the fallacy of **Illicit Conversion**

Obversion

- **Obversion** requires two steps: (1) changing the quality (without changing the quantity), and (2) replacing the predicate with its term complement.
- ***class complement*** is the group consisting of everything outside the class. For example, the complement of the class of dogs is the group that includes everything that is not a dog (cats, fish, trees, and so on).
- The **term complement** is the word or group of words that denotes the class complement. For terms consisting of a single word, the term complement is usually formed by simply attaching the prefix "non" to the term.

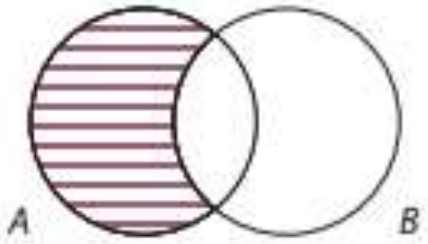
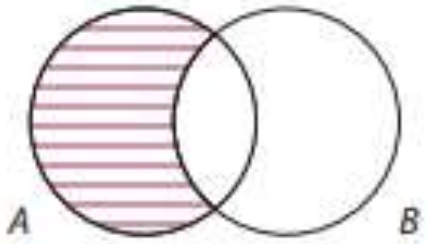
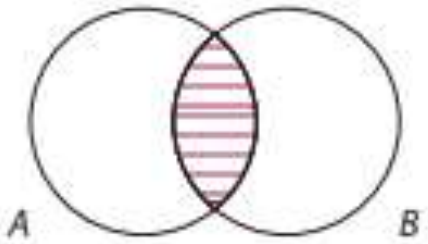
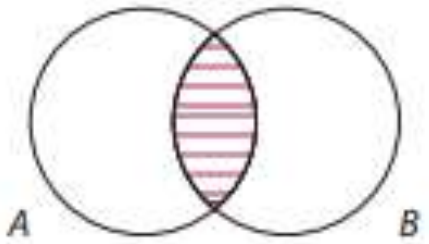
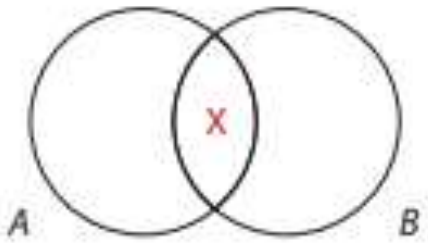
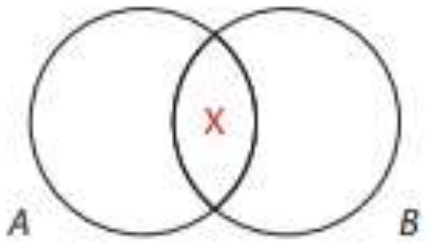
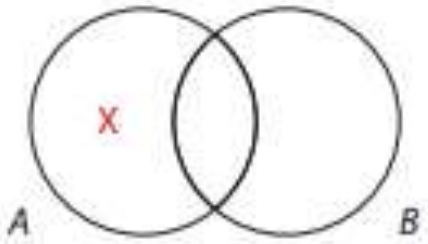
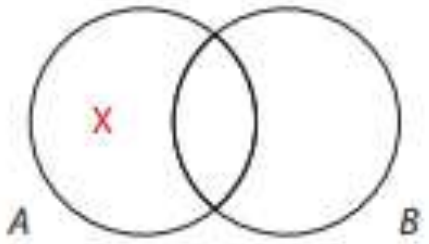
Examples:

All horses are animals. _____

Some horses are animals. _____

No horses are animals. _____

Some horses are not animals. _____

Given statement form	Obverse
All *A* are *B*. 	No *A* are non-*B*. 
No *A* are *B*. 	All *A* are non-*B*. 
Some *A* are *B*. 	Some *A* are not non-*B*. 
Some *A* are not *B*. 	Some *A* are non-*B*. 

The diagram for each given statement form is identical to the diagram for its obverse. This means that each of the four types of categorical proposition is logically equivalent to (and has the same meaning as) its obverse.

Using Obversion to evaluate the validity of immediate inferences.

Examples:

1. All **cats** are **dogs**. Therefore, no cats are **non-dogs**.
2. No **cats** are **dogs**. Therefore, all cats are **non-dogs**.
3. Some **cats** are **dogs**. Therefore, some **cats** are not **non-dogs**.
4. Some **cats** are not **dogs**. Therefore, some **cats** are **non-dogs**.

Contraposition

- **Contraposition** requires two steps:

(1) switching the subject and predicate terms and

(2) replacing the subject and predicate terms with their term complements.

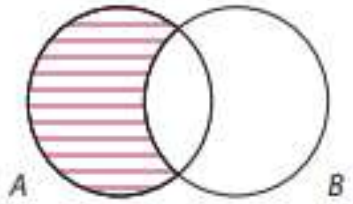
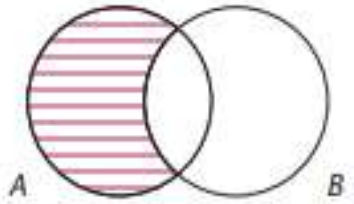
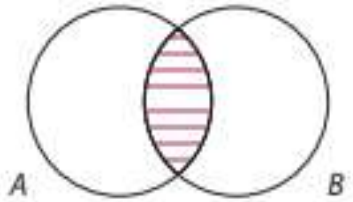
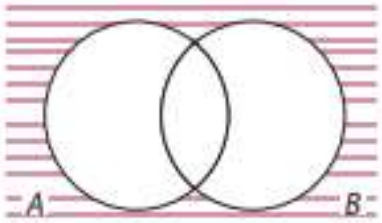
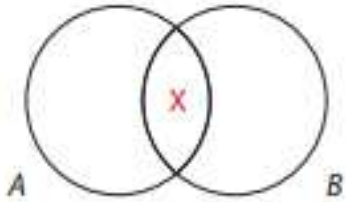
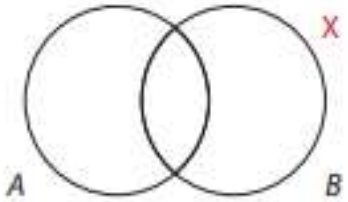
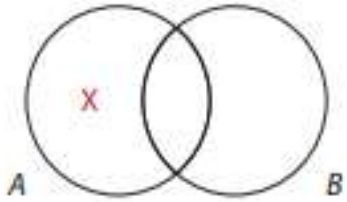
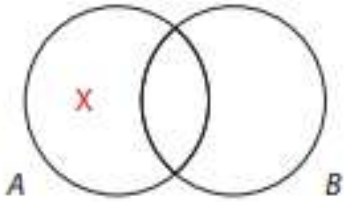
So,

All **A** are **B** → All **non-B** are **non-A**.

No **A** are **B** → No **non-B** are **non-A**.

Some **A** are **B** → Some **non-B** are **non-A**.

Some **A** are not **B** → Some **non-B** are not **non-A**.

Given statement form	Contrapositive
All *A* are *B*. 	All non-*B* are non-*A*. 
No *A* are *B*. 	No non-*B* are non-*A*. 
Some *A* are *B*. 	Some non-*B* are non-*A*. 
Some *A* are not *B*. 	Some non-*B* are not non-*A*. 

- The diagrams of **A** and **O** statements reveal that they are identical, logically equivalent (and have the same meaning)to the diagrams of their contrapositive.
- The diagrams of the **E** and **I** statements are neither identical to nor the exact opposite of the diagrams of their contrapositives. This means that contraposing an **E** or **I** statement gives a new statement whose truth value is logically undetermined in relation to the given statement.

Contraposition can be used to evaluate the validity of immediate inference.

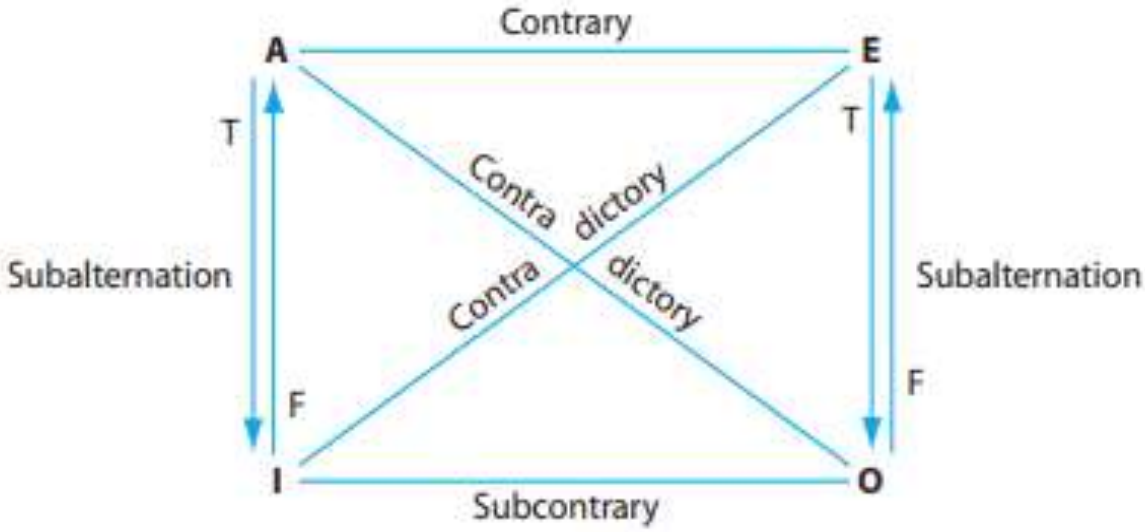
Examples:

1. All dogs are cats. So, all non- cats are non- dogs.
2. No dogs are cats. So, no non-cats are non-dogs.
3. Some dogs are cats. So, some non-cats are non-dogs.
4. Some dogs are not cats. So, some non-cats are not non-dogs.

- **E** and **I** commit the fallacy of **Illicit Contraposition**

6.5 The Traditional Square of Opposition

• It is an arrangement of lines that illustrates logically necessary relations among the four kinds of categorical propositions.



•The four relations in the traditional square of opposition may be characterized as follows:

Contradictory = opposite truth value (It expresses complete opposition between propositions.)

Contrary = at least one is false (not both true)

Subcontrary = at least one is true (not both false)

Subalteration = truth flows downward, falsity flows upward

Testing Immediate Inferences

Examples:

1. All Swiss watches are true works of art.
   Therefore, it is false that no Swiss watches are true works of art.

2. Some viruses are structures that attack T cells.
   Therefore, some viruses are not structures that attack T cells

Illicit Contrary

3. It is false that all *A* are *B.*
   Therefore, no *A* are *B.*

4. It is false that no *A* are *B.*
   Therefore, all *A* are *B.*

Illicit Subcontrary

5.Some *A* are *B.*
Therefore, it is false that some *A* are not *B.*

6.Some *A* are not *B.*
   Therefore, some *A* are *B.*

Illicit Subalternation

7. Some *A* are not *B.*
   Therefore, no *A* are *B.*

8. It is false that all *A* are *B.*
Therefore, it is false that some *A* are *B.*

Conditionally valid

9. All *A* are *B.*
Therefore, it is false that no *A* are *B.*

TSO and existential fallacy

- All witches who fly on broomsticks are fearless women. Therefore, some witches who fly on broomsticks are fearless women.
- No wizards with magical powers are malevolent beings. Therefore, it is false that all wizards with magical powers are malevolent beings.